

Nakul Upadhyaya

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Education

University of Toronto (St. George) | Toronto, ON, CA
PhD in Mechanical and Industrial Engineering

Aug. 2022 – Current (Est. 2027)

- **Advisor:** Eldan Cohen (OptiMaL Lab @ UofT)
- **Thesis:** Shape Generalized Trees: Foundations and Induction Algorithms
- **Research Areas:** Explainable AI, Model-Based Interpretability, Deep Learning,
- **Awards:** Mitacs Accelerate (\$10,000), 2026 SGS Conference Grant (\$1,300), 2024 MIE Grad. Research Symposium 1st Place Poster Presentation (\$500), 2026 MIE Grad. Research Symposium - 3rd Place Oral Presentation (\$200)

Purdue University | West Lafayette, IN, US
Bachelors of Science in Industrial Engineering

Aug. 2018 – May 2022

Publications

Shape Generalized Trees

- **Upadhyaya, N.**, Cohen, E. (2026). Literati: Towards Anytime Optimal Shape Generalized Trees via AO* *40th Conference on Neural Information Processing Systems*. [paper]
- **Upadhyaya, N.**, Cohen, E. (2025). Empowering Decision Trees via Shape Function Branching. *39th Conference on Neural Information Processing Systems*. [paper] [code]

Generalized Additive Models

- **Upadhyaya, N.**, Ling, Z. Aoyama, T. Yano, K. Cohen, E. (2026). Interpretable Prediction of Product Quality Via Sparse Multi-Task Additive Models. *Preprint Submitted to Computers and Industrial Engineering*
- **Upadhyaya, N.**, Cohen, E. (2024). NeurCAM: Interpretable Neural Clustering via Additive Models. *27th European Conference on Artificial Intelligence*. [paper] [code]

Other

- Song, Q. C., Shin, H. J., **Upadhyaya, N.**, & Teo, T. (2023). Technology and Measurement in Asia. *Technology and Measurement around the Globe*. Cambridge University Press. [paper]

Professional Experience

GEMINI Medicine | Toronto, ON, CA
PhD Researcher - Industry Partner

March 2024 - Current

- Developed machine learning models to predict delirium in hospitalized patients using retrospective data from over 5,000+ patients, achieving high performance with only 15 clinical features chosen through iterative feature selection.
- Performed fairness analysis across various Ontario Marginalization Index axes, mitigating equalized-odds differences between advantaged and disadvantaged subgroups.

Konica Minolta | Toronto, ON, CA
PhD Researcher - Industry Partner

Aug. 2022 - June 2025

- Developed a full end-to-end quality prediction system using multivariate sensor time-series data, including a novel multi-task Neural GAM architecture for ordinal defect forecasting.
- Built a correlation-graph-based feature reduction pipeline that removed redundant sensors and improved both model sparsity and interpretability.
- Designed and deployed interactive analytics dashboards that delivered real-time predictions and feature-level shape-function explanations to manufacturing engineers.

Kohl's Technology | Menomonee Falls, WI, USA
Data Scientist (Intern)

May 2022 - Aug. 2022

- Collaborated with the procurement and supply chain teams to develop integer programs for inventory allocation across all 512 Kohl's stores.
- Designed an automated customer segmentation engine leveraging T-SNE and Mini-Batch K-Means to provide valuable insights to the sales and marketing divisions.
- Deployed machine learning and optimization algorithms using Google Cloud Platform services such as Vertex AI and BigQuery.

Whirlpool Corporation | Benton Harbor, MI, USA
Supply Chain Analyst (Intern)

May 2021 - Aug. 2021

- Led requirement gathering sessions with cross-functional teams of process owners for Lodestar – the new manufacturing execution system responsible for managing the production of over 25 million units a year
- Developed and implemented data visualization dashboards displaying scheduling and yield information for work center team leaders across all 9 Whirlpool plants in North America

Teaching Experience

Data Science Institute | Toronto, ON, CA
Technical Facilitator - Deep Learning

January 2026 - Current

- Designed and developed curriculum and instructional materials for the Deep Learning module of the DSI MicroCertificates Program, spanning foundational to state-of-the-art topics.
- Delivered lectures on core deep learning concepts including optimization and learning theory (ex. gradient descent, regularization), computer vision architectures (ex. YOLO, Vision Transformers), and large language model architectures (ex. BERT, GPT).
- Coordinated a team of teaching assistants to ensure smooth course delivery, including assignment grading, student support, and logistics.

Other Instructional Experience

Department of Mechanical and Industrial Engineering | Toronto, ON, CA
University of Toronto

- MIE 370 - Introduction to Machine Learning | Tutorial TA | Fall 2026
- MIE 223 - Data Science | Tutorial TA | Winter 2025/2026
- MIE 350 - Design and Analysis of Information Systems | Lab TA | Fall 2022/2023
- MIE 253/353 - Data Modeling | Lab TA | Winter 2023, Fall 2024/2025

Data Science Institute | Toronto, ON, CA
University of Toronto

- DSI-CARTE Machine Learning Bootcamp | Lab TA | Summer 2023-2025, Fall 2023
- DSI-CARTE Post-Doctoral Deep Learning Bootcamp | Lab TA | Summer 2023

Centre for Analytics & AI Engineering (CARTE) | Toronto, ON, CA
University of Toronto

- M-Lab (Mitsubishi) AI Workshop | Lab Instructor & Content Developer | Fall 2025
- Enbridge AI Bootcamp | Lab Instructor & Content Developer | Fall 2023

Academic Service

- NeurIPS 2026 Evaluations and Datasets Track Reviewer